

The Complete Residual and a Fictional Closing Mechanism

Recovered mathematics, complete conditional deduction, and analytical tests

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Updated October 5, 2026

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About this continuation

This focused extract contains the complete newly documented continuation from the lifetime-census companion. It preserves that document's section numbers 52-73 so that every equation and proof can be located in the complete record. References to sections 1-51 refer to the accompanying full companion. The earlier reconstruction estimate is restated in section 52; its prior proof is preserved in section 42 of the companion.

Start with sections 68-73 for the current direct calculation: joint boundary cancellation, exact prime jumps, complete finite expansion, a proved small-insertion allowance, and positive prime-square credits. Sections 59-60 define the residual. Section 61 gives the earlier fictional mechanism and sections 63-66 its tests. Sections 53-57 preserve the supplied compression and positive-allowance proofs. Status ledgers are in sections 67 and 73.

The growth law F1 remains explicitly hypothetical. The direct work in sections 68-73 uses the actual multiplicative identities and does not assume F1. The complete signed lower bound remains the arithmetic target. Numerical calculations throughout are diagnostic only.

52. Recovery of the unsaved continuation and its dictionary

This part incorporates the compression and fixed-cutoff positive allowance supplied in the conversation, together with the subsequent complete prime-discrepancy residual. It supersedes their historical “not saved” notices. Sections 1-51 are preserved. The estimate required at the end remains a one-sided estimate for the complete signed expression.

All logarithms are natural. Let

$$\mu(n) = \begin{cases} (-1)^r & n \text{ is a product of } r \text{ distinct primes,} \\ 0 & n \text{ has a squared prime factor,} \end{cases} \quad M(x) = \sum_{n \leq x} \mu(n), \quad \mathcal{A}(x) = \int_1^x \frac{M(t)}{t} dt.$$

Set $U(X) = \mathcal{A}(2X) - \mathcal{A}(X)$, and

$$\Theta(z) = \sum_{\substack{m \leq z \\ m \text{ modd}}} \log(z/m), \quad e(z) = \Theta(z) - z/2 + (\log 2)/2.$$

Here e is the odd-counting remainder, not the earlier prime-discrepancy function $\psi(x) - x$. For the cutoffs below, every argument of e is at least one. The earlier odd-counting estimate is $|e(z)| \leq 1/(4z)$.

For an integer $N = 2X$, let

$$L = \lfloor \sqrt{N} \rfloor, \quad 1 \leq H \leq L/4, \quad K = L - H, \quad m_K = \sum_{a \leq K} \frac{\mu(a)}{a},$$

and define

$$Q_K(N) = \sum_{a, b \leq K} \mu(a)\mu(b)\Theta\left(\frac{N}{ab}\right), \quad G_K(N) = \sum_{\substack{a, b > K \\ ab \leq N}} \mu(a)\mu(b) \log \frac{N}{ab}.$$

The already established reconstruction, including its cutoff tail, is

$$\boxed{U(N/2) = 2(\log 2)M(K) - Q_K(N) + G_K(N), \quad |G_K(N)| \leq 6H^3/L.} \quad (\text{C1})$$

For large L , choose $H = \lfloor L^{2/3} \rfloor$; finitely many smaller cutoffs can use any admissible H , and are absorbed in constants. Then $G_K = O(\sqrt{N})$. The full center is

$$Q_K(N) = \frac{N}{2} m_K^2 - \frac{\log 2}{2} M(K)^2 + \sum_{a, b \leq K} \mu(a)\mu(b) e\left(\frac{N}{ab}\right). \quad (\text{C2})$$

The second term has a minus sign and is retained throughout.

The comparison notation used later is distinct: $F(x, y)$ is an actual Möbius sum over integers with no prime factor at most y ; $\mathfrak{F}(x, y)$ is a separately defined continuous comparison. Their difference produces $\mathcal{Z}_N$. None of G_K , F , $\mathfrak{F}$, and $\mathcal{Z}_N$ denotes the historical energy I_M .

53. Reciprocal-preserving compression with both centers exact

Let $\eta = N^{-1/4}$, and choose integer nodes

$$t_0 = 1, \quad t_{j+1} = \min\{K, t_j + \max(1, \lfloor \eta t_j \rfloor)\}, \quad t_J = K. \quad (\text{C3})$$

For an integer a strictly between successive nodes $t < r$, put

$$\vartheta_a = \frac{1/t - 1/a}{1/t - 1/r}.$$

Assign $(1 - \vartheta_a)\mu(a)$ to t , and $\vartheta_a\mu(a)$ to r . At a node assign its coefficient to that node once. Let β_j be the resulting signed masses. These assignments give, exactly,

$$\sum_j \beta_j = M(K), \quad \sum_j \frac{\beta_j}{t_j} = m_K, \quad \sum_j \frac{|\beta_j|}{t_j} \leq \sum_{a \leq K} \frac{|\mu(a)|}{a} \leq 1 + \log K. \quad (\text{C4})$$

The masses need not have absolute value at most one. Define

$$\widehat{Q}_K(N) = \sum_{i,j} \beta_i \beta_j \Theta\left(\frac{N}{t_i t_j}\right).$$

Then

$$|Q_K(N) - \widehat{Q}_K(N)| \leq \eta^2 N (1 + \log K)^2 = \sqrt{N} (1 + \log K)^2. \quad (\text{C5})$$

In particular,

$$\widehat{Q}_K(N) = \frac{N}{2} m_K^2 - \frac{\log 2}{2} M(K)^2 + \sum_{i,j} \beta_i \beta_j e\left(\frac{N}{t_i t_j}\right). \quad (\text{C6})$$

No centering approximation occurs.

Proof of the error bound. Fix $b \leq K$, allowing b to be a node, and put $c = N/b$. In logarithmic coordinates,

$$k(x) = \Theta(c e^{-x}) = \sum_{m \text{ odd}} (\log(c/m) - x)_+.$$

Its second derivative as a measure is a sum of unit atoms at the hinge locations $x = \log(c/m)$. On a node cell $[t, r]$, write $g = r - t$, $d = \log(r/t)$, and $v = \log(a/t)$. A cell of width one has no interior integer to move. On every other cell, $g \leq \eta t$.

First use the log-linear weight v/d . A single hinge in the cell contributes at most $d/4$ to the interpolation error at any integer. Its total contribution over the cell is at most

$$gd/4 \leq g^2/(4t) \leq \eta^2 t/4 \leq \eta^2 c/(4m).$$

Each hinge belongs to one cell, with boundary hinges giving no error. The relevant odd indices lie between c/K and c . Since $c/K \geq 1$, their reciprocal sum is at most $1 + \log K$. Thus the sum of log-linear errors is at most $\eta^2 c(1 + \log K)/4$.

Next replace the log-linear weight by the reciprocal weight. Concavity of $1 - e^{-v}$ and its second derivative bound give

$$0 \leq \vartheta_a - v/d \leq \frac{d^2}{8(1 - e^{-d})}.$$

The endpoint kernel difference is at most $c(1 - e^{-d})/t$, because the number of odd integers up to a positive real z is at most z when it is nonzero. Consequently this change costs at most

$$\frac{cd^2}{8t} \leq \frac{(1 + \eta)\eta^2 c}{8a}.$$

Using $\eta \leq 1$, the complete one-coordinate error, summed over the original coefficients, is at most

$$\frac{\eta^2 N}{2b} (1 + \log K).$$

Telescope the two-coordinate replacement. In the first coordinate sum with the original weights $|\mu(b)|/b$; in the second use the compressed weights $|\beta_j|/t_j$. Formula (C4) bounds both by $1 + \log K$. This proves (C5), including every cross-term.

There are $O(\eta^{-1}(1 + \log K))$ nodes. Below $2/\eta$ there are at most $2/\eta$ integers. Above it, the node increment is at least $\eta t/2$, so geometric growth gives the claimed count.

54. The compressed coordinates are historical energy means

Define

$$\Delta_j = \frac{1}{t_j} - \frac{1}{t_{j+1}}, \quad c_j = \frac{1}{\Delta_j} \int_{t_j}^{t_{j+1}} \frac{M(u)}{u^2} du \quad (0 \leq j < J). \quad (C7)$$

Integration by parts on each cell yields

$$\beta_0 = c_0, \quad \beta_j = c_j - c_{j-1} \quad (0 < j < J), \quad \beta_J = M(K) - c_{J-1}. \quad (C8)$$

For example the contribution of an interior coefficient $\mu(a)$ to c_j is $(1/a - 1/t_{j+1})/\Delta_j = 1 - \vartheta_a$. The differences in (C8) therefore reproduce exactly the assignment in section 53. The final coefficient explicitly retains the inherited terminal.

Partial summation and the weighted variance identity give

$$m_K = \frac{M(K)}{K} + \sum_j \Delta_j c_j, \quad \boxed{I_M(K) = \sum_j \Delta_j c_j^2 + V, \quad 0 \leq V \leq \frac{1}{4}.} \quad (C9)$$

Here

$$V = \sum_j \int_{t_j}^{t_{j+1}} \frac{(M(u) - c_j)^2}{u^2} du.$$

A width-one cell has zero variance because M is constant almost everywhere there. On a cell of width $g \geq 2$, the range of M is at most g . A variable supported in an interval of length g has variance at most $g^2/4$: expand the nonnegative product $(Y - \min Y)(\max Y - Y)$, then complete the square in the mean. Hence that cell contributes at most

$$\Delta_j g^2/4 = \frac{g^3}{4t_j t_{j+1}} \leq \eta^2 g/4.$$

Summing gives $V \leq \eta^2(K-1)/4 \leq 1/4$. Thus the omitted within-cell energy has a fixed allowance; it is not another unbounded remainder.

Combining (C1) and (C5) gives the complete bridge

$$\boxed{|U(N/2) - [2(\log 2)M(K) - \widehat{Q}_K(N)]| \leq 6H^3/L + \sqrt{N}(1 + \log K)^2.} \quad (C10)$$

The sufficient arithmetic statement remains $\widehat{Q}_K(N) \geq -C_\varepsilon N^{1/2+\varepsilon}$ for every $\varepsilon > 0$, with constants independent of N .

55. Exact output grouping and its cutoff hypotheses

This records the arithmetic grouping from the supplied continuation without extending it to arbitrary cutoffs. Assume $N \geq 16$, $K \geq 2$, and

$$2N/3 \leq K^2 \leq N.$$

Let

$$T(z) = \log_+ z - 2\log_+(z/2) + \log_+(z/4), \quad P(N) = \sum_{p \text{ modd prime}} T(N/p). \quad (C11)$$

Thus $T(z) = \log z$ for $1 < z \leq 2$, $T(z) = \log(4/z)$ for $2 \leq z < 4$, and it is zero otherwise. In particular $T \geq 0$.

Write $Q_K(N) = \sum_{n \leq N} c_K(n) \log(N/n)$. Ordinary Dirichlet convolution gives

$$c_K(n) = 2\mu(n)\mathbf{1}_{n \leq K} - 2\mathbf{1}_{2|n}\mu(n/2)\mathbf{1}_{n \leq 2K} - [\mu(n) - \mathbf{1}_{2|n}\mu(n/2)] + g_K(n), \quad g_K(n) = \sum_{\substack{ab=n \\ a, b > K}} \mu(a)\mu(b). \quad (\text{C12})$$

Indeed $\mu * \mathbf{1}_{\text{odd}} = \delta_1 - \delta_2$. Expanding the two truncated factors gives (C12). The tail has no further odd multiplier: $a, b > K$ and an odd multiplier at least three would force $n > 3K^2 > N$.

In the tail, $\mu(a)\mu(b) \neq 0$ requires prime exponents at most two. Suppose the odd part of n has exactly one prime p appearing to the first power, with a nontrivial squared odd factor. In a contributing factorization, squared factors belong to both factors. The ratio of the unequal factors is at least p , or at least $p/2$ when the exponent of two is one. Thus that ratio is at least $3/2$. But both factors exceeding K would imply $n > (3/2)K^2 \geq N$. The tail vanishes; all the other terms in (C12) vanish because of the squared odd factor. Higher exponents vanish automatically.

If the odd part is a square, the same argument excludes an odd exponent of two in a tail term (ratio two). Actual square outputs contribute

$$S_K(N) = \sum_{K < a \leq \sqrt{N}} \mu(a)^2 \log(N/a^2) \geq 0.$$

For odd-prime outputs, grouping $p, 2p, 4p$ gives the tent credit $P(N)$. The inherited prefix in (C12) gives $-2(\log 2)\pi_{\text{odd}}(K/2, K]$ for those outputs. Therefore

$$\boxed{Q_K(N) = P(N) + S_K(N) - 2(\log 2)\pi_{\text{odd}}(K/2, K] + R_{\geq 2}(N).} \quad (\text{C13})$$

Here $R_{\geq 2}$ is defined to be the complete contribution of the other outputs whose odd part has at least two first-power primes. Their inherited terms remain included. This is an exact grouping, not positivity of $R_{\geq 2}$.

56. A proved positive allowance at each fixed roughness cutoff

Let $P^-(1) = \infty$ and

$$F(x, y) = \sum_{P^-(n) > y} \mu(n)T(x/n), \quad B_\theta(N) = -F(N, N^\theta), \quad 0 < \theta < 1. \quad (\text{C14})$$

The summation includes the unit $n = 1$. For every **fixed** θ ,

$$\boxed{B_\theta(N) = [\rho(1/\theta - 1) + o_\theta(1)] \frac{N}{4 \log N} > 0 \quad (N \text{ sufficiently large}),} \quad (\text{C15})$$

where

$$\rho(v) = 1 \quad (0 \leq v \leq 1), \quad v\rho'(v) = -\rho(v-1) \quad (v > 1). \quad (\text{C16})$$

All allowable numbers of distinct prime factors, with alternating signs, occur in (C15).

Here is the analytic derivation, including the cutoff qualification. Factoring each nonunit squarefree integer by its smallest prime gives the universally valid identity

$$\boxed{F(x, y) = T(x) - \sum_{p > y} F(x/p, p).} \quad (\text{C17})$$

An exact separated version is

$$F(x, y) = T(x) - \sum_{p > y} T(x/p) - \sum_{y < p \leq \sqrt{x}} \{F(x/p, p) - T(x/p)\}. \quad (\text{C18})$$

For $x \geq 16$, $2 \leq y < x/4$, this simplifies to

$$F(x, y) = -P(x) - \sum_{y < p \leq \sqrt{x}} F(x/p, p). \quad (\text{C19})$$

The conditions in (C19) matter. For example one must not drop the unit term at a small child argument. Equation (C17) is the formula used in subsequent exact recursions.

The prime number theorem and partial summation imply

$$P(x) \sim \frac{x}{\log x} \int_{1/4}^1 T(1/t) dt = \frac{x}{4 \log x}. \quad (\text{C20})$$

For completeness the integral equals $\int_{1/4}^{1/2} \log(4t) dt + \int_{1/2}^1 \log(1/t) dt = 1/4$.

An absolute estimate controls transition strips in the recursion. For fixed $V > 1$, uniformly in $y \geq x^{1/V}$,

$$\sum_{P^-(n) > y} |\mu(n)| T(x/n) \ll_V x / \log x \quad (x \text{ large}). \quad (\text{C21})$$

A nonzero summand has $x/4 < n < x$ and at most $\lfloor V \rfloor$ prime factors. If there are j factors, choose the largest p , and write $n = dp$. Then $p \geq (x/4)^{1/j}$, while $p \leq x/d$; the elementary consequence $\pi(z) \ll z / \log z$ of PNT bounds the number of such choices by $O_j(x/(d \log x))$. The reciprocal sum for the remaining distinct prime factors is bounded by

$$\frac{1}{(j-1)!} \left(\sum_{y < p \leq x} \frac{1}{p} \right)^{j-1} \ll_V 1.$$

The last bound also follows by prime partial summation. Sum over the finitely many j .

Put $u = \log x / \log y$, and write the leading term as $-F(x, x^{1/u}) \sim f(u)x/(4 \log x)$. For $1 < u \leq 2$, only singleton primes contribute for large x , so $f(u) = 1$. Induct on bounded ranges of u . In (C19) the child ratio is $w = \log(x/p) / \log p \in [1, u-1]$. On $w \geq 1 + \gamma$ the inductive asymptotic is uniform on compact intervals. On the remaining transition strip (C21) and prime partial summation give $O_V(\gamma x / \log x)$. Applying prime partial summation on the main range, then taking $x \rightarrow \infty$ and $\gamma \downarrow 0$, gives

$$f(u) = 1 - \int_1^{u-1} \frac{f(w)}{w} dw. \quad (\text{C22})$$

This is exactly the defining delay equation for $f(u) = \rho(u-1)$. The same induction gives the asserted compact uniformity away from $u = 1$. Finally

$$v\rho(v) = \int_{v-1}^v \rho(t) dt \quad (v \geq 1) \quad (\text{C23})$$

follows by differentiation and its value at one. A first zero of ρ would contradict (C23), proving strict positivity. These steps establish (C15) for each fixed θ ; they do not assert an error estimate uniform as $\theta \rightarrow 0$.

For $\theta = 1/3$, no product of three permitted primes fits below N , and

$$B_{1/3}(N) = P(N) - \sum_{N^{1/3} < p < q} T(N/(pq)) \sim (1 - \log 2) \frac{N}{4 \log N}. \quad (\text{C24})$$

The semiprime-to-prime ratio is $\int_{1/3}^{1/2} [\alpha(1-\alpha)]^{-1} d\alpha = \log 2 < 1$. Thus the prime allowance pays for that whole family and leaves a positive surplus.

57. The smaller-prime contribution is the same whole comparison

For $N \geq 4$ and $N^\theta \geq 2$, define

$$D_\theta(N) = - \sum_{2 < p \leq N^\theta} F(N/p, p).$$

The least-prime partition gives

$$D_\theta(N) - B_\theta(N) = F(N, 2).$$

The latter is exactly $U(N/2)$. To verify this directly, put $A_o(x) = \sum_{n \leq x, n \text{ modd}} \mu(n) \log(x/n)$. The even squarefree terms give $\mathcal{A}(x) = A_o(x) - A_o(x/2)$, and the tent gives

$$F(N, 2) = A_o(N) - 2A_o(N/2) + A_o(N/4) = \mathcal{A}(N) - \mathcal{A}(N/2). \quad (\text{C25})$$

Consequently

$$\boxed{Q_K(N) = 2(\log 2)M(K) + B_\theta(N) - D_\theta(N) + G_K(N).} \quad (\text{C26})$$

The lower bound for Q_K , or its compressed version with (C5), is therefore equivalent up to the paid errors to

$$D_\theta(N) \leq B_\theta(N) + C_\varepsilon N^{1/2+\varepsilon}. \quad (\text{C27})$$

The positivity of B_θ proves a credit, not (C27). All smaller-prime branches must remain coupled to that credit. This is the exact unresolved comparison stated in the user's checkpoint.

58. A historical-coordinate identity that retains interactions

This auxiliary identity gives another way to inspect the compressed object; it is not an additional closing requirement. Set $c_J = M(K)$, so that (C8) expresses β as successive differences. Write $W_{ij} = \Theta(N/(t_i t_j))$. Summation by parts in both indices yields

$$\widehat{Q}_K = c^\top A c,$$

where, for $i, j < J$,

$$A_{ij} = W_{ij} - W_{i+1,j} - W_{i,j+1} + W_{i+1,j+1}, \quad A_{iJ} = W_{iJ} - W_{i+1,J}, \quad A_{JJ} = W_{JJ}.$$

All these entries are nonnegative: each log-coordinate hinge has a nonnegative mixed finite difference, and the endpoint differences are nonnegative. This is entrywise positivity, not positive semidefiniteness. The row sums are

$$r_i = \Theta(N/t_i) - \Theta(N/t_{i+1}) \quad (i < J), \quad r_J = \Theta(N/K).$$

Thus the exact polarization identity is

$$\boxed{\widehat{Q}_K = \sum_i r_i c_i^2 - \sum_{i < j} A_{ij} (c_i - c_j)^2.} \quad (\text{C28})$$

Both the positive term and the debit are determined by the same historical coordinates. Positivity of the coefficients alone cannot compare their sizes. Formula (C28) retains the full terminal and every off-diagonal interaction.

Background for the standard notation and PNT: NIST Digital Library of Mathematical Functions, section 27.2, <https://dlmf.nist.gov/27.2>. Background for the Dickman function: A. Hildebrand and G. Tenenbaum, Integers without large prime factors, Journal de Theorie des Nombres de Bordeaux 5 (1993), 411-484, section 2, https://jtnb.centre-mersenne.org/item/JTNB_1993__5_2_411_0.pdf. The interpolation, output, and comparison arguments above are supplied here explicitly; these background references are not claims of external validation of the new estimates.

59. The exact continuous comparison

Let ρ be the function defined by

$$\rho(v) = 1 \quad (0 \leq v \leq 1), \quad v\rho'(v) = -\rho(v-1) \quad (v > 1).$$

Define a DIFFERENT function, with a continuous prime-density measure,

$$\mathfrak{F}(x, y) = T(x) - \int_y^\infty T(x/r) \frac{\rho(\log r / \log y - 1)}{\log r} dr, \quad y \geq 2. \quad (\text{Z3})$$

The integral is finite, because $T(x/r) = 0$ when $r \geq x$. Formula (Z3) is a definition, not an approximation asserted for the actual primes.

This comparison satisfies the exact continuous recursion

$$\mathfrak{F}(x, y) = T(x) - \int_y^\infty \mathfrak{F}(x/t, t) \frac{dt}{\log t}. \quad (\text{Z4})$$

Here is a direct derivation. Iterating the integral operator gives ordered factors $y < t_1 < \dots < t_k$. Symmetry converts this to

$$\mathfrak{F}(x, y) = T(x) + \sum_{k \geq 1} \frac{(-1)^k}{k!} \int_{t_1, \dots, t_k > y} T\left(\frac{x}{t_1 \dots t_k}\right) \prod_{j=1}^k \frac{dt_j}{\log t_j}. \quad (\text{Z5})$$

At each fixed x, y , only finitely many k can contribute, since a nonzero term requires $t_1 \dots t_k < x$.

To evaluate the product density, write $t_j = y^{u_j}$, $u_j > 1$. Apart from the common factor $y^{u_1 + \dots + u_k}$, the measure is the product of du_j/u_j . Put

$$h(u) = \frac{\mathbf{1}_{\{u > 1\}}}{u}, \quad \sigma(u) = \sum_{k \geq 1} \frac{(-1)^k}{k!} h^{*k}(u).$$

The sum is locally finite. Multiplying a convolution by the sum of its arguments gives

$$u\sigma(u) = - \left[1 + \int_0^{u-1} \sigma(v) dv \right] \quad (u > 1),$$

while $\sigma = 0$ on $(0, 1)$. Therefore $1 + \int_0^u \sigma(v) dv = \rho(u)$, so $\sigma = \rho'$. Changing back from the product coordinate to r gives precisely (Z3). The ordered-factor construction proves (Z4).

The identity

$$v\rho(v) = \int_{v-1}^v \rho(w) dw \quad (v \geq 1) \quad (\text{Z6})$$

follows by differentiation and the value at $v = 1$. A first zero is impossible, so $\rho > 0$; its differential equation then gives monotonicity.

For each fixed $0 < \theta < 1$, partial rescaling of (Z3) gives

$$-\mathfrak{F}(N, N^\theta) \sim \rho(1/\theta - 1) \frac{N}{4 \log N}. \quad (\text{Z7})$$

Thus this comparison has exactly the leading coefficient from the earlier fixed- θ result. Extending the definition to $y = 2$ does not extend that asymptotic statement for the actual arithmetic function F ; their difference is retained explicitly next.

Bounded contribution at the full cutoff

For $N \geq 4$, $T(N) = 0$, hence

$$\boxed{\mathfrak{F}(N, 2) \leq 0.} \quad (\text{Z8})$$

It is also uniformly bounded. From (Z6) and monotonicity, $\rho(m) \leq \rho(m-1)/m$, so $\rho(v) \leq 1/\lfloor v \rfloor!$ for $v \geq 1$. For $N \geq 16$, set $v_N = \log N / \log 2 - 3$ and $m = \lfloor v_N \rfloor$. The tent restricts r to $(N/4, N)$, giving

$$\begin{aligned} 0 \leq -\mathfrak{F}(N, 2) &\leq (\log 2) \frac{N}{\log(N/4)} \rho(v_N) \\ &\leq \frac{N}{2m!} < 8 \frac{2^m}{m!} \leq 8e^2. \end{aligned}$$

For $4 \leq N < 16$, a direct integral majorant gives a bound of 16. In particular,

$$\boxed{0 \leq -\mathfrak{F}(N, 2) \leq 16e^2 \quad (N \geq 4).} \quad (\text{Z9})$$

This is an all-scale bound for the explicitly defined comparison function, not for U .

60. The complete prime-discrepancy residual

For a two-variable function H , write

$$(\mathsf{P}H)(x, y) = \sum_{p > y} H(x/p, p), \quad (\mathsf{C}H)(x, y) = \int_y^\infty H(x/t, t) \frac{dt}{\log t}.$$

Equations (C17) and (Z4) say $(I + \mathsf{P})F = T$ and $(I + \mathsf{C})\mathfrak{F} = T$. Subtracting gives

$$(I + \mathsf{P})(F - \mathfrak{F}) = -(\mathsf{P} - \mathsf{C})\mathfrak{F}. \quad (\text{Z10})$$

At each finite x, y , the inverse on the functions here is a finite alternating expansion: once the product of the successive factors exceeds x , the tent support makes the term zero. Thus no convergence assumption or infinite resolvent is being imposed.

Define the prime-counting discrepancy for $t \geq 2$ by

$$\Delta_\pi(t) = \pi(t) - 1 - \int_2^t \frac{dv}{\log v}.$$

Its Stieltjes measure on $(2, \infty)$ is the actual odd-prime counting measure minus $dt/\log t$. For $P^+(1) = 1$, put

$$\boxed{\mathcal{K}_N(t) = \sum_{\substack{d \leq N/t \text{ odd} \\ P^+(d) < t}} \mu(d) \mathfrak{F}\left(\frac{N}{dt}, t\right).} \quad (\text{Z11})$$

The strict inequality $P^+(d) < t$ matters at a prime atom: it includes only smaller actual primes in the inherited cofactor. The $d = 1$ term is included.

Then the complete error is

$$\boxed{\mathcal{Z}_N = \int_{(2, N]} \mathcal{K}_N(t) d\Delta_\pi(t), \quad F(N, 2) = \mathfrak{F}(N, 2) - \mathcal{Z}_N.} \quad (\text{Z12})$$

Proof. Let $R = (\mathsf{P} - \mathsf{C})\mathfrak{F}$. Expanding (Z10) gives

$$F - \mathfrak{F} = - \sum_{k \geq 0} (-\mathsf{P})^k R.$$

An increasing chain of actual primes has one squarefree product d and coefficient $\mu(d)$. In the R term, the next factor t exceeds every prime in d . Writing R as an integral against $d\Delta_\pi$ and interchanging the finite sum with that integral gives (Z11)-(Z12). Every inherited cofactor occurs once. The continuous factors remaining above t are already contained in $\mathfrak{F}$.

This is a hybrid ordering identity: the primes below the discrepancy insertion are actual, while the factors above it belong to the explicitly defined comparison. The sum accounts for every possible position of that insertion. It is not a claim that the actual primes are independently distributed.

Return to the original lower bound

Combining (C25) and (Z12),

$$\boxed{B_\theta(N) - D_\theta(N) = -\mathfrak{F}(N, 2) + \mathcal{Z}_N.} \quad (\text{Z13})$$

The entire θ -dependent partition has recombined. The prime contribution has been used once, and no subset has been separately discarded.

Since (Z9) is bounded independently of N , the original one-sided comparison is equivalent, up to constants, to

$$\boxed{\mathcal{Z}_N \geq -C_\varepsilon N^{1/2+\varepsilon} \quad \text{for every } \varepsilon > 0.} \quad (\text{Z14})$$

For the raw quadratic form, the inherited identity now reads

$$Q_K(N) = 2(\log 2)M(K) - \mathfrak{F}(N, 2) + \mathcal{Z}_N + G_K(N). \quad (\text{Z15})$$

The already-established compression error transfers this exact expression to $\widehat{Q}_K$. The original centering is retained through this equality; it has not been independently estimated or assigned a favorable sign.

61. A fictional closing mechanism for the exact residual

Status of this section. The finite identities below are real identities. The growth law labeled F1 is an invented hypothesis for exploration, not an established property of the primes. The deduction from F1 is a rigorous conditional proof. The tests in sections 63-66 examine the proposed mechanism and state its limitations. F1 is a sufficient certificate, not a new necessary condition added to the original one-sided target.

Use the exact $\mathcal{K}_N$, Δ_π , and $\mathcal{Z}_N$ from section 60. For real $N \geq 4$, partition $(2, N]$ into the disjoint intervals

$$I_{N,j} = (\max\{2, N/2^{j+1}\}, N/2^j], \quad 0 \leq j < J_N, \quad J_N = \lceil \log_2(N/2) \rceil.$$

Define the complete signed contribution of each interval and its squared sum by

$$z_{N,j} = \int_{I_{N,j}} \mathcal{K}_N(t) d\Delta_\pi(t), \quad S_N = \sum_{j=0}^{J_N-1} z_{N,j}^2. \quad (\text{F0})$$

All inherited cofactors and continuous branches are combined inside $z_{N,j}$ before it is squared. End-point atoms belong to exactly one interval. In particular $\mathcal{Z}_N = \sum_j z_{N,j}$ exactly. This does not assert that contributions in different intervals are independent.

For an integer $r \geq 0$, put

$$E_r = \sup_{4 \leq N \leq 2^{r+2}} \frac{S_N}{N}.$$

This is a finite quantity for each r . On a bounded N -range there are finitely many arithmetic factors, the comparison integrals are bounded, and all kernels have bounded finite sums. The supremum includes real cutoffs and intermediate values, not only powers of two.

The fictional law is:

$$1 + E_{r+1} \leq \left(1 + \frac{A}{r+1}\right) (1 + E_r) \quad \text{for every } r \geq 0, \quad (\text{F1})$$

for one finite constant $A \geq 0$, independent of r . It says that, after dividing the squared contribution by the counting scale N , the additional fractional cost of another doubling is at most $A/(r+1)$. Raw energy need not decrease, primes need not be regularly spaced, and individual signed contributions may be negative. The decreasing quantity in this fictional rule is the permitted fractional increase of the normalized running budget.

Conditional theorem. If F1 holds for the actual coefficients in section 60, then

$$|\mathcal{Z}_N| \ll_A \sqrt{N} (1 + \log N)^{(A+1)/2}. \quad (\text{F2})$$

Consequently $\mathcal{Z}_N \geq -C_\varepsilon N^{1/2+\varepsilon}$ for every $\varepsilon > 0$, and the selected analytic closing argument applies.

Proof. Multiplying F1 and using $\log(1+x) \leq x$ gives

$$1 + E_r \leq (1 + E_0) \prod_{k=1}^r (1 + A/k) \leq (1 + E_0) e^{A(1+\log r)} \ll_A (1+r)^A.$$

Choose r with $N \leq 2^{r+2}$ and $r \ll 1 + \log N$. Then $S_N \ll_A N(1 + \log N)^A$. Cauchy-Schwarz gives

$$\mathcal{Z}_N^2 = \left(\sum_j z_{N,j} \right)^2 \leq J_N S_N \ll_A N(1 + \log N)^{A+1}.$$

Taking square roots proves (F2). Every fixed power of $\log N$ is $O_\varepsilon(N^\varepsilon)$, which gives the required one-sided bound. No independence, positivity of each interval, or numerical extrapolation enters this deduction.

The sole invented arithmetic step in this proof is F1. The product calculation and Cauchy-Schwarz cannot prove F1. An argument that says “the continuous comparison decays, therefore F1 holds” would leave precisely that step unjustified.

62. What the fictional proof would complete

This section spells out the closing implications so that the arithmetic premise and the analytic conclusion have no missing link. It also applies if the original one-sided bound is proved directly, without F1.

Assume $\mathcal{Z}_N \geq -C_\varepsilon N^{1/2+\varepsilon}$ for every $\varepsilon > 0$. Equations (Z9) and (Z12) give

$$U(N/2) = \mathfrak{F}(N, 2) - \mathcal{Z}_N \leq C_\varepsilon N^{1/2+\varepsilon}. \quad (\text{F3})$$

This is the needed upper bound; a lower bound on U is not needed. If it is known initially only for integer N , it extends to real arguments. Indeed, almost everywhere,

$$U'(X) = \frac{M(2X) - M(X)}{X}, \quad |U'(X)| \leq 1 + 1/X \leq 2 \quad (X \geq 1).$$

Adjacent half-integer sample arguments therefore differ by at most a fixed constant.

Fix $q > 1/2$. Choose $\varepsilon < q - 1/2$, telescope the dyadic increments U , and absorb the bounded initial interval to obtain

$$\mathcal{A}(x) \leq C_q x^q \quad (x \geq 1). \quad (\text{F4})$$

Increase C_q if necessary so that $g_q(x) = C_q x^q - \mathcal{A}(x) \geq 0$. For $\Re s > \max(1, q)$, absolute convergence and two partial integrations give

$$\int_1^\infty \mathcal{A}(x) x^{-s-1} dx = \frac{1}{s^2 \zeta(s)}, \quad \int_1^\infty g_q(x) x^{-s-1} dx = \frac{C_q}{s-q} - \frac{1}{s^2 \zeta(s)}. \quad (\text{F5})$$

For clarity, the positivity principle needed here has an elementary analytic proof. A Mellin transform of a nonnegative locally integrable function with a finite real abscissa of convergence cannot be analytic at that abscissa. Otherwise take a real point just to its right inside an analytic disk. The Taylor derivatives are the moments of powers of $-\log x$. At a real point slightly to the left, their Taylor terms all have the same sign. Tonelli’s theorem identifies the convergent Taylor sum with the integral there, contradicting the definition of the abscissa. This is the usual Landau positivity argument.

Apply that principle to g_q , with $1/2 < q < 1$. The right side of (F5) is analytic in a neighborhood of every real $s > q$: $\zeta(s) > 0$ for real $s > 1$, and it is nonzero for $0 < s < 1$ because the positive alternating eta series equals $(1 - 2^{1-s})\zeta(s)$. At $s = 1$, $1/\zeta(s)$ is analytic with a zero. Therefore the abscissa of the nonnegative integral is at most q . Its integral is holomorphic on $\Re s > q$, and (F5) continues $1/\zeta(s)$ there holomorphically. A zero of ζ in that half-plane would produce a pole, a contradiction.

Letting $q \downarrow 1/2$ excludes zeros to the right of the critical line. The functional equation reflects non-trivial zeros under $s \mapsto 1 - s$, excluding those to the left as well. This is the conditional RH conclusion. The proof uses the original one-sided residual bound; it requires no separate historical-energy, Fourier-tail, or variational-regularity estimate afterward.

63. Analytical attack I: positivity fails, and the upper band is explicit

The fictional proof deliberately allows negative contributions. An unconditional claim $\mathcal{Z}_N \geq 0$ at every cutoff is already false.

At $N = 4$, the actual tent sum is $F(4, 2) = -\log(4/3)$. For $2 < r < 4$, the Dickman factor in (Z3) is one, so

$$\mathcal{Z}_4 = \log(4/3) - \int_2^4 \frac{\log(4/r)}{\log r} dr. \quad (\text{F6})$$

The integrand is $\log 4 / \log r - 1$, a convex function on $[2, 4]$. Jensen's inequality gives

$$\int_2^4 \frac{\log(4/r)}{\log r} dr \geq 2 \left(\frac{\log 4}{\log 3} - 1 \right) > \log(4/3),$$

the last inequality involving fixed elementary constants. Thus $\mathcal{Z}_4 < 0$ analytically. This refutes universal positivity; it does not contradict a negative allowance of square-root size.

There is also a useful exact simplification, valid for all $N \geq 4$:

$$\boxed{\mathcal{K}_N(t) = \log(N/t) \quad (N/2 < t \leq N).} \quad (\text{F7})$$

Only $d = 1$ can occur in that interval, and $N/t \leq 2 \leq t$, so the continuous child integral vanishes. Consequently

$$\boxed{z_{N,0} = \sum_{N/2 < p \leq N} \log(N/p) - \int_{N/2}^N \frac{\log(N/t)}{\log t} dt.} \quad (\text{F8})$$

The top interval already contains a definite smoothed prime-counting error. It has no remaining complicated cofactor to remove.

More generally, when $t > \sqrt{N}$, every allowed cofactor satisfies $d < t$ and every continuous child has argument below its cutoff. Hence

$$\boxed{\mathcal{K}_N(t) = F(N/t, 2) \quad (t > \sqrt{N}).} \quad (\text{F9})$$

This exposes an exact smaller-scale Möbius response inside the larger-prime discrepancy integral. It is a useful structural identity, not by itself an upper or lower bound.

64. Analytical attack II: how strong the invented law really is

F1 implies a square-root estimate, up to logarithms, for each individual $z_{N,j}$, because $|z_{N,j}| \leq \sqrt{S_N}$. In particular it would bound (F8) at that scale. The prime number theorem used for (C15) supplies only an asymptotic error $o(N/\log N)$ for this fixed smooth prime weight; that statement does not imply a square-root error. Thus F1 cannot be deduced merely by repeating the PNT partial summation already used for the positive allowance.

The square budget is stronger than the original objective in a second way. It bounds $|\mathcal{Z}_N|$, whereas the closing argument needs only its negative part. It also limits every interval before cancellations between intervals are used. For an abstract two-interval vector $(L, -L)$, the total is zero but its squared sum is $2L^2$. This is an algebraic illustration of the difference between the two criteria, not a proposed model of the actual primes.

Consequently a failure to prove F1 would not introduce a new gap in the established route. One can instead try to prove the original signed lower bound directly, allowing a large negative interval to be paid by an actual positive interval elsewhere. The current paper retains those cross-interval credits in $\mathcal{Z}_N$; none is thrown away in (Z12) or (Z15).

An absolute-value shortcut is also too costly with the elementary estimates available here. Since $0 \leq \rho \leq 1$,

$$|\mathfrak{F}(x, y)| \leq \log 2 + (\log 2)x / \log y.$$

Summing this bound over all allowed d , even after discarding zero coefficients, gives the crude estimate

$$|\mathcal{K}_N(t)| \ll \frac{N}{t}(1 + \log N), \quad 2 < t \leq N.$$

Integrating against the total variation of $d\Delta_\pi$, and using only the harmonic bound for the prime sum, yields

$$\int_{(2,N]} |\mathcal{K}_N(t)| d|\Delta_\pi|(t) \ll N(1 + \log N)^2. \quad (\text{F10})$$

This is an upper bound too large for the desired scale, not a proof that every possible absolute estimate must be large. It pinpoints what the straightforward triangle inequality fails to accomplish.

The remaining analytic safeguards are equally concrete. The fixed- θ coefficient in (C15) cannot be substituted at $\theta = \log 2 / \log N$ without a new uniform error estimate. The continuous comparison at $y = 2$ is defined exactly and bounded, but its arithmetic difference is still $\mathcal{Z}_N$. Finally the interval contributions share the same primes and inherited Möbius coefficients; calling them independent or orthogonal would be an extra unsupported assumption. The proof of F2 uses neither claim.

65. Numerical attacks on signs, normalization, and reconstruction

The following calculations are floating-point diagnostics. They check formulas, cutoff conventions, and plausible failure modes. They are not interval-certified estimates, a proof of F1, or evidence that a bound holds at untested scales. No fitted exponent is extrapolated.

The saved script constructs ρ by the method of steps for its differential equation and evaluates the continuous comparison by adaptive quadrature. It computes each dyadic interval contribution from its prime sum minus its continuous integral, retaining $P^+(d) < t$. A separate direct odd Möbius sum computes $F(N, 2)$. The check is the independent agreement of $\sum_j z_{N,j}$ with $\mathfrak{F}(N, 2) - F(N, 2)$.

N	$\mathcal{Z}_N$, numerical	S_N/N , numerical	Sum of negative interval parts
4	-0.377360	0.035600	0.377360
8	-0.198395	0.003114	0.198395
16	0.702913	0.098857	0.472937
32	1.207232	0.042517	0.247214
64	1.312628	0.030819	0.761474
128	1.486366	0.025758	0.673683
256	1.142616	0.031613	2.321994

The largest reconstruction discrepancy in this run was less than $2 \cdot 10^{-9}$. In particular there are negative interval contributions even when the total is positive. The sampled normalized square sum is not monotonically decreasing: it rises sharply from $N = 8$ to $N = 16$. Therefore the simplest “normalized energy always decreases” guess is contradicted by this diagnostic. F1 was formulated with a running supremum and an allowed fractional increase, not with monotone decay of S_N/N .

The table samples only seven endpoints. It does not evaluate the supremum over real N in E_r , and must not be described as a verification of F1.

A second saved script checks the compression and historical-coordinate formulas at $N = 256, 1024, 4096, 16384$:

N	K	Nodes	$ Q_K - \widehat{Q}_K $	Within-cell variance V
256	12	10	0.048493	0.004132
1024	24	17	0.151295	0.010078
4096	49	27	0.181035	0.013893
16384	103	43	0.123568	0.010196

The mass and historical-coordinate identities agreed to approximately 10^{-15} ; the largest raw reconstruction discrepancy was below $5 \cdot 10^{-11}$. The stated error allowances held at these samples. Their proof is the analytic argument in sections 53-54, not this computation.

The accompanying source package's diagnostics/ folder contains:

- residual_diagnostics.py and residual_diagnostics.json;
- compression_diagnostics.py and compression_diagnostics.json.

The scripts record the numerical method and preserve the individual interval contributions.

66. What survived the attacks, and the precise next arithmetic object

The fictional mechanism survives as a valid conditional implication. Its arithmetic law has not been derived. The attacks establish the following distinctions.

Proposed step	Finding
Every residual is nonnegative	False: section 63 proves $\mathcal{Z}_4 < 0$
The normalized square sum always decreases	The diagnostic has a large increase between 8 and 16
F1 implies the desired rate	Proved conditionally in section 61
The continuous comparison bounds the actual residual automatically	No: (Z12) keeps their difference explicitly
Fixed-cutoff PNT asymptotics imply F1	No such implication follows; even the top interval needs a stronger error estimate
The exact formula loses a terminal or prime atom	The finite expansion retains both; independent numerical checks agree
Failure of the square-budget route creates another mandatory gate	No: the original one-sided signed target remains sufficient

The strongest exact new simplification from this audit is (F9), together with its explicit top interval (F8). At a dyadic cutoff, the lower-bound comparison can now be inspected as

$$\mathcal{Z}_N = \underbrace{\int_{(N/2, N]} \log(N/t) d\Delta_\pi(t)}_{\text{explicit top contribution}} + \underbrace{\int_{(2, N/2]} \mathcal{K}_N(t) d\Delta_\pi(t)}_{\text{all inherited remaining contributions}} . \quad (\text{F11})$$

This equality is valid at every real $N \geq 4$, not only dyadic cutoffs. No estimate for one term has been subtracted from another budget. The desired lower bound concerns their combined signed value. A proof can use compensation between them, instead of proving the stronger individual estimates that F1 would entail.

The unresolved real statement is still exactly

$$\boxed{\int_{(2, N]} \mathcal{K}_N(t) d\Delta_\pi(t) \geq -C_\varepsilon N^{1/2+\varepsilon} \text{ for every } \varepsilon > 0.} \quad (\text{F12})$$

Here all coefficients, cutoffs, and continuous comparison terms are defined in sections 52 and 59-60. The fictional growth law explains one sufficient shape of a proof. It has not supplied the arithmetic justification of that law.

67. Documentation map and status ledger

This edition records every displayed link in the supplied continuation and the subsequent exact residual. The former disconnected-workspace statements are historical notices, now superseded for this material.

Supplied or subsequent result	Location in this edition	Status
Complete strip removal and inherited terminal	(C1), section 52; original proof in section 42	Proved identity and allowance
Reciprocal-preserving masses and both centers	(C3)-(C6), section 53	Proved
Compression error with all cross-terms	(C5), section 53	Proved
Historical means, terminal, and $V \leq 1/4$	(C7)-(C10), section 54	Proved
Squarefull cancellation, square credit, prime credit	(C12)-(C13), section 55	Proved under displayed cutoff conditions
Positive fixed- θ allowance and $1 - \log 2$ case	(C15), (C24), section 56	Proved for fixed θ
Complete smaller-prime comparison	(C25)-(C27), section 57	Exact; its growth estimate remains open here
Historical-coordinate polarization	(C28), section 58	Exact; no PSD assertion
Continuous comparison and full-cutoff bound	(Z3)-(Z9), section 59	Proved for the comparison function
Complete actual-prime residual	(Z10)-(Z15), section 60	Exact
Fictional shrinking fractional budget	(F1), section 61	Invented hypothesis
Deduction from that budget to closure	(F2)-(F5), sections 61-62	Proved conditional implication
Negative residual and explicit larger-prime response	(F6)-(F9), section 63	Proved
Finite computational checks	Section 65 and saved scripts/results	Diagnostics only
Required all-scale signed lower bound	(F12), section 66	Not proved by this continuation

The original sections and their historical indexing remain available. The source package preserves the previous edition, both previously saved residual notes, the new continuation files, numerical diagnostics, the build scripts, and a file manifest with checksums. The lifetime companion is separate from the 12-page summary, the running proof manuscript, and the larger research master; this update does not silently replace those other documents.

68. Joint transfer: regularity, traces, and the common boundary

This continuation performs the joint integration by parts proposed after section 67. It uses the actual residual (Z12). The fictional growth law F1 is not assumed. The result is an exact finite expansion, an unconditional allowance for all sufficiently small discrepancy insertions, and explicit positive prime-square contributions. The complete remaining signed lower bound is kept as the same arithmetic target.

To keep the dictionary unambiguous, write

$$\nu(t) = \pi(t) - 1, \quad \ell(t) = \int_2^t \frac{dv}{\log v}, \quad D(t) = \nu(t) - \ell(t) = \Delta_\pi(t) \quad (t \geq 2). \quad (\text{J1})$$

Thus ν counts odd primes and $D(2) = 0$. The symbol $P(N)$ used for the earlier weighted prime allowance is unchanged; it is not used for this counting function.

Define the logarithmic derivative of the tent almost everywhere by

$$h(x) = xT'(x) = \mathbf{1}_{1 < x < 2} - \mathbf{1}_{2 < x < 4}.$$

Values at its three endpoints can be assigned arbitrarily in ordinary integrals. Let

$$\mathfrak{H}(x, y) = x \partial_x \mathfrak{F}(x, y) = h(x) - \int_y^\infty h(x/v) \frac{\rho(\log v / \log y - 1)}{\log v} dv. \quad (\text{J2})$$

For each integer $r \geq 0$, define a left-continuous kernel and its response:

$$\begin{aligned} \mathcal{K}_{N,r}^-(t) &= \sum_{\substack{d \text{ modd} \\ P^+(d) < t}} \mu(d) \mathfrak{F}\left(\frac{N}{dt^{r+1}}, t\right), \\ \mathcal{H}_{N,r}(t) &= \sum_{\substack{d \text{ modd} \\ P^+(d) < t}} \mu(d) \mathfrak{H}\left(\frac{N}{dt^{r+1}}, t\right). \end{aligned} \quad (\text{J3})$$

Every sum is finite: its summands vanish when $d \geq N/t^{r+1}$. The unit cofactor $d = 1$ is included. Let $\mathcal{K}_{N,r}^+$ use $P^+(d) \leq t$ instead; it is the right-continuous version. In particular $\mathcal{K}_{N,0}^- = \mathcal{K}_N$, and the prime atoms in the original residual use the left version.

These kernels are of bounded variation on each bounded interval in t . Here are the relevant regularity facts. The tent is Lipschitz in $\log x$. Its integrals in the finite ordered-factor formula (Z5) are locally absolutely continuous in their parameters. Differentiation in x under those finite integrals gives (J2), and differentiating the lower endpoint in (Z4) gives

$$\partial_y \mathfrak{F}(x, y) = \frac{\mathfrak{F}(x/y, y)}{\log y}. \quad (\text{J4})$$

On a bounded N, t -range only finitely many squarefree cofactors occur. A summand entering or leaving at $N/(dt^{r+1}) = 1$ has value zero. The tent corners cause derivative discontinuities, not jumps of the kernel. The only kernel jumps are therefore the changes in the finite cofactor set at primes. These facts justify the Stieltjes product rules used below without a further regularity premise.

At $R = \sqrt{N}$, both kernel traces agree and

$$\mathcal{K}_{N,0}^-(R) = \mathcal{K}_{N,0}^+(R) = F(R, 2). \quad (\text{J5})$$

Indeed $\mathfrak{F}(R/d, R) = T(R/d)$. Any cofactor excluded by strictness at $d = R$ has zero tent weight. For $t > R$, the already established identity is $\mathcal{K}_{N,0}^-(t) = F(N/t, 2)$, with

$$\frac{d}{dt} F(N/t, 2) = -\frac{M(N/t) - M(N/(2t))}{t} \quad \text{almost everywhere.} \quad (\text{J6})$$

Here $M(x) = 0$ for $0 < x < 1$.

69. The boundary cancels and every prime jump descends exactly

Factor a cofactor newly admitted at an odd prime p as $d = pa$, with $P^+(a) < p$. The nonzero coefficients are squarefree, so $\mu(pa) = -\mu(a)$. This gives

$$\boxed{\mathcal{K}_{N,r}^+(p) - \mathcal{K}_{N,r}^-(p) = -\mathcal{K}_{N,r+1}^-(p).} \quad (\text{J7})$$

For $r = 0$, the right side is $-\mathcal{K}_{N/p}^-(p)$. The jump vanishes when $p^{r+2} \geq N$. Thus the first jumps stop at the square-root boundary, the next family stops at the cube-root boundary, and subsequent families stop at the corresponding prime-power boundary.

Between primes, (J2)-(J4) give

$$\frac{d}{dt} \mathcal{K}_{N,r}(t) = -\frac{r+1}{t} \mathcal{H}_{N,r}(t) + \frac{\mathcal{K}_{N,r+1}^-(t)}{\log t}.$$

Combining this with the jumps proves the exact signed-measure identity

$$\boxed{d\mathcal{K}_{N,r}^+(t) = -(r+1)\mathcal{H}_{N,r}(t)\frac{dt}{t} - \mathcal{K}_{N,r+1}^-(t) dD(t).} \quad (\text{J8})$$

All terms on the right use the same actual prime-discrepancy measure.

For right-continuous bounded-variation functions f, g , the product rule in the convention needed here is

$$d(fg) = f_- dg + g df, \quad (\text{J9})$$

where g is evaluated after a jump in the second integral. At a jump this identity is simply $f_- \Delta g + g \Delta f = fg - f_- g_-$. This convention accounts for the interaction at a shared prime atom exactly once.

Apply (J9) below and above $R = \sqrt{N}$. The low piece is

$$\int_{(2,R]} \mathcal{K}_{N,0}^- dD = F(R, 2)D(R) - \int_{(2,R]} D d\mathcal{K}_{N,0}^+.$$

The high piece is

$$\int_{(R,N]} \mathcal{K}_{N,0}^- dD = -F(R, 2)D(R) + \int_R^N D(t) \frac{M(N/t) - M(N/(2t))}{t} dt.$$

The displayed boundary terms cancel, including when R itself is prime, because (J7) gives a zero jump there. Using (J8), the resulting whole identity is

$$\boxed{\mathcal{Z}_N = \int_2^N D(t) \mathcal{H}_{N,0}(t) \frac{dt}{t} + \int_{(2,N]} D(t) \mathcal{K}_{N,1}^-(t) dD(t).} \quad (\text{J10})$$

At a prime in the last integral, $D(t)$ is the post-jump value and the kernel is the pre-jump value. The inherited correction is not omitted. It is the term that will generate the squared discrepancy together with the actual prime count.

70. An exact finite expansion retaining coincident prime interactions

The expansion can be based at any Y with $2 \leq Y < N$. Define

$$\nu_Y(t) = \pi(t) - \pi(Y), \quad \ell_Y(t) = \int_Y^t \frac{dv}{\log v}, \quad D_Y(t) = D(t) - D(Y) = \nu_Y(t) - \ell_Y(t).$$

Introduce coefficients by the convergent generating function near $z = 0$:

$$\boxed{\sum_{k \geq 0} \mathcal{B}_{k,Y}(t) z^k = e^{-\ell_Y(t)z} (1-z)^{-\nu_Y(t)}} \quad (\text{J11})$$

These coefficients are distinct from the earlier fixed- θ allowance B_θ . They are explicit polynomials in the actual count and its smooth centering. The first terms are

$$\begin{aligned} \mathcal{B}_{0,Y} &= 1, & \mathcal{B}_{1,Y} &= D_Y, \\ \mathcal{B}_{2,Y} &= \frac{1}{2}(D_Y^2 + \nu_Y), \\ \mathcal{B}_{3,Y} &= \frac{1}{6}(D_Y^3 + 3\nu_Y D_Y + 2\nu_Y^2). \end{aligned} \quad (\text{J12})$$

They can also be calculated by

$$k\mathcal{B}_{k,Y} = D_Y \mathcal{B}_{k-1,Y} + \nu_Y \sum_{j=0}^{k-2} \mathcal{B}_{j,Y},$$

with an empty sum for $k = 1$.

Exact differential identity. With post-jump values on the right,

$$\boxed{d\mathcal{B}_{k+1,Y}(t) = \mathcal{B}_{k,Y}(t) dD(t)}. \quad (\text{J13})$$

Between primes the generating function differentiates to $-z/\log t$ times itself. At a prime it is multiplied by $(1-z)^{-1}$, so the jump of its $(k+1)$ -st coefficient equals the post-jump k -th coefficient. This proves both parts of (J13). In particular replacing $\mathcal{B}_{2,Y}$ by $D_Y^2/2$ would lose the $\nu_Y/2$ contribution from coincident jumps.

For $q_Y = \lceil \log N / \log Y \rceil$, define

$$J_r(Y) = \int_{(Y,N]} \mathcal{B}_{r,Y}(t) \mathcal{K}_{N,r}^-(t) dD(t).$$

Using (J9), (J13), and then (J8), we obtain

$$J_r(Y) = (r+1) \int_Y^N \mathcal{B}_{r+1,Y}(t) \mathcal{H}_{N,r}(t) \frac{dt}{t} + J_{r+1}(Y). \quad (\text{J14})$$

There is no omitted boundary term: $\mathcal{B}_{r+1,Y}(Y) = 0$, while $\mathcal{K}_{N,r}^+(N) = 0$. When $r = q_Y - 1$, $N/t^{r+1} < 1$ on $(Y, N]$, so the kernel and $J_r(Y)$ vanish. Iteration is finite, yielding

$$\boxed{\int_{(Y,N]} \mathcal{K}_N(t) dD(t) = \int_Y^N \sum_{k=1}^{q_Y-1} k\mathcal{B}_{k,Y}(t) \mathcal{H}_{N,k-1}(t) \frac{dt}{t}}. \quad (\text{J15})$$

Every inherited prime-power interaction appears once. The terms remain in one signed sum and share the measure dt/t . This is a finite identity for each N, Y , without a limiting exchange or a growth assumption.

71. An unconditional allowance for the complete small-insertion range

Let

$$\mathfrak{e}_{N,Y} = \int_{(2,Y]} \mathcal{K}_N(t) dD(t), \quad 2 \leq Y \leq N.$$

This interval is indexed by the location of the discrepancy insertion. All actual cofactors below that insertion and all continuous factors above it remain present. It is a different partition from the integer-output family $C_y(N)$ in the preserved small-prime-output note.

Uniform finite bound. For every such N, Y ,

$$|\mathfrak{e}_{N,Y}| \leq [\nu(Y) + \ell(Y)] 2^{\nu(Y)} (\log 2 + Y^2 e^Y). \quad (\text{J16})$$

In particular, for sufficiently large N , set

$$Y_N^{(0)} = \frac{1}{2} \log N.$$

Then

$$\mathfrak{e}_{N,Y_N^{(0)}} = O_\varepsilon(N^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0. \quad (\text{J17})$$

Equivalently its absolute value is at most $N^{1/2+o(1)}$. Smaller N can use a cutoff of two, with finitely many initial values absorbed in the constants. A sharper cutoff is obtained below.

Proof. The factorial estimate for ρ from section 59 gives, for $y \geq 2$,

$$\begin{aligned} \int_y^\infty \frac{\rho(\log v / \log y - 1)}{\log v} dv &\leq \sum_{m \geq 0} \frac{y^{m+2}}{m! \log y} \\ &= \frac{y^2 e^y}{\log y}. \end{aligned}$$

Here the m -th interval is $[y^{m+1}, y^{m+2}]$, and $\rho \leq 1/m!$ there. Since $|T| \leq \log 2$,

$$|\mathfrak{F}(x, y)| \leq (\log 2) \left(1 + \frac{y^2 e^y}{\log y} \right) \quad \text{uniformly for every } x > 0. \quad (\text{J18})$$

There are exactly $2^{\nu(t-)}$ odd squarefree products of the primes below t , before imposing any size cutoff. Every nonzero coefficient in $\mathcal{K}_N(t)$ belongs to this finite family. Therefore, for $2 < t \leq Y$,

$$|\mathcal{K}_N(t)| \leq 2^{\nu(Y)} (\log 2 + Y^2 e^Y).$$

The total variation of dD on $(2, Y]$ is $\nu(Y) + \ell(Y)$: its positive part consists of the prime atoms and its negative part is the continuous density. This proves (J16).

Only the elementary upper estimate $\pi(Y) \ll Y / \log Y$ is needed for its rate. One proof uses the product of the primes in $(n, 2n]$, which divides $\binom{2n}{n} < 4^n$. Thus their count is at most $n \log 4 / \log n$. Split $(\sqrt{Y}, Y]$ into dyadic intervals and bound primes below $\sqrt{Y}$ trivially; this gives the stated upper estimate. Splitting the integral for $\ell(Y)$ at $\sqrt{Y}$ similarly gives $\ell(Y) \ll Y / \log Y$. The logarithm of the right side of (J16) is consequently

$$Y + O\left(\frac{Y}{\log Y} + \log Y\right).$$

Substituting $Y = \frac{1}{2} \log N$ proves (J17). This estimate holds for the entire insertion interval, including arbitrarily large inherited products; it is not a termwise truncation of those products.

Sharper complete allowance. The factorial bound can be improved by retaining the exact Laplace moment of ρ . Put

$$L_\rho(a) = \int_0^\infty \rho(u) e^{au} du, \quad J(a) = \int_0^a \frac{e^v - 1}{v} dv \quad (a \geq 0).$$

The singularity of the integrand at zero is removable. The factorial estimate just used proves convergence of all these moments and justifies differentiation and integration by parts. Extend $\rho(u) = 0$ for $u < 0$. The delay equation gives $(u\rho(u))' = \rho(u) - \rho(u-1)$. Consequently

$$aL'_\rho(a) = (e^a - 1)L_\rho(a), \quad L_\rho(a) = L_\rho(0)e^{J(a)} \leq e e^{J(a)}.$$

The last inequality follows from $L_\rho(0) \leq \sum_{m \geq 0} 1/m! = e$. Change variables $v = y^{u+1}$ in the comparison integral to obtain the stronger uniform bound

$$\begin{aligned} \int_y^\infty \frac{\rho(\log v / \log y - 1)}{\log v} dv &= y \int_0^\infty \frac{\rho(u)y^u}{u+1} du \\ &\leq yL_\rho(\log y). \end{aligned}$$

Thus another explicit, all- N bound for the same complete insertion interval is

$$|\epsilon_{N,Y}| \leq [\nu(Y) + \ell(Y)] 2^{\nu(Y)} (\log 2) [1 + eY e^{J(\log Y)}]. \quad (\text{J18a})$$

Elementary l'Hopital comparison gives $J(\log Y) \sim Y / \log Y$. The prime number theorem, already used in section 56, gives $\nu(Y) \sim Y / \log Y$. Taking logarithms in (J18a) therefore shows

$$\log(\text{right side of (J18a)}) \leq (1 + \log 2 + o(1)) \frac{Y}{\log Y}.$$

For the remainder of this continuation choose, for large N ,

$$Y_N = \frac{1}{4} \log N \log \log N.$$

Then

$$|\epsilon_{N,Y_N}| \leq N^{(1+\log 2)/4+o(1)} = O(\sqrt{N}). \quad (\text{J18b})$$

The exponent $(1+\log 2)/4$ is strictly less than $1/2$, so the last big-O assertion follows for all sufficiently large N . Set $Y_N = 2$ on the finitely many smaller values if needed. This sharper allowance is unconditional and uses no hypothesis on $D(t)$.

Combine (J15) with this paid interval, using the coefficients based at the same Y_N :

$$\mathcal{Z}_N = \epsilon_{N,Y_N} + \int_{Y_N}^N \sum_{k=1}^{q_{Y_N}-1} k \mathcal{B}_{k,Y_N}(t) \mathcal{H}_{N,k-1}(t) \frac{dt}{t}. \quad (\text{J19})$$

Here $q_{Y_N} = O(\log N / \log \log N)$. Starting the coefficient recursion at Y_N is essential: all $\mathcal{B}_{k,Y_N}(Y_N)$, $k \geq 1$, vanish. One must not simply delete the lower interval from the expansion based at two, because that would leave its nonzero boundary terms unpaid.

72. Positive even coefficients and complete prime-square credits

The coefficients in (J11) contain exact positivity. For every $j \geq 0$,

$$\mathcal{B}_{2j,Y}(t) \geq 0 \quad (t \geq Y). \quad (\text{J20})$$

To prove this, put $m = \nu_Y(t)$, $a = \ell_Y(t)$. If $m = 0$, the coefficient is $(-a)^k/k!$. If $m \geq 1$, the elementary integral for $(1-z)^{-m}$ gives

$$\mathcal{B}_{k,Y}(t) = \frac{1}{k!(m-1)!} \int_0^\infty (u-a)^k u^{m-1} e^{-u} du. \quad (\text{J21})$$

Expansion of $e^{z(u-a)}$, or differentiation at zero, proves the identity. For even k the integrand is nonnegative. This is a deterministic representation of coefficients of the exact counting function; it assumes no randomness or independence of primes.

For the second level, the complete coefficient is

$$2\mathcal{B}_{2,Y} = D_Y^2 + \nu_Y. \quad (\text{J22})$$

On the terminal interval $\sqrt{N/2} < t < \sqrt{N}$, with $t \geq 2$, only the unit cofactor contributes to $\mathcal{H}_{N,1}$, its continuous child integral vanishes, and its tent response is one. Thus $\mathcal{H}_{N,1}(t) = 1$ there. If $Y \leq \sqrt{N/2}$, its exact nonnegative contribution is

$$\mathcal{C}_{2,Y}(N) = \int_{\sqrt{N/2}}^{\sqrt{N}} ([D(t) - D(Y)]^2 + \pi(t) - \pi(Y)) \frac{dt}{t} \geq 0. \quad (\text{J23})$$

It contains the actual prime-counting discrepancy inside the same square. For the cutoff Y_N , prime partial summation gives the explicit lower allowance

$$\mathcal{C}_{2,Y_N}(N) \geq [2(1 - 1/\sqrt{2}) + o(1)] \frac{\sqrt{N}}{\log N}. \quad (\text{J24})$$

Indeed one can omit the nonnegative square and apply the prime number theorem to the remaining count integral; $\pi(Y_N)$ gives a negligible subtraction.

There is a second useful grouping that pays both sides of the unit-cofactor tent. Decompose $\mathcal{H}_{N,1}$ into its unit term $h(N/t^2)$ and all remaining terms from (J2)-(J3). Isolate only the $\nu_Y(t)h(N/t^2)$ part of (J22). If $Y \leq \sqrt{N/2}$, integration by parts gives

$$\begin{aligned} \int_Y^N \nu_Y(t) h(N/t^2) \frac{dt}{t} &= \frac{1}{2} \int_{(Y,N]} T(N/t^2) d\pi(t) \\ &= \boxed{\frac{1}{2} \sum_{p>Y} T(N/p^2) \geq 0.} \end{aligned} \quad (\text{J25})$$

There are no end terms because the tent vanishes at both ends of its support. This combines its positive and negative derivative portions before assigning the credit. The sum already has the allowed size:

$$0 \leq \frac{1}{2} \sum_{p>Y_N} T(N/p^2) \ll \frac{\sqrt{N}}{\log N},$$

and more precisely

$$\frac{1}{2} \sum_{p>Y_N} T(N/p^2) \sim (3 - 2\sqrt{2}) \frac{\sqrt{N}}{\log N}. \quad (\text{J26})$$

For the constant, scale $p = \sqrt{N}u$ and calculate $\int_{1/2}^1 T(1/u^2) du = 3 - 2\sqrt{2}$. Prime partial summation then proves (J26). The upper estimate follows already from the elementary prime-counting bound used in section 71.

Equations (J23) and (J25) are two different groupings of overlapping parts of the second level. Their credits must not be added as if they came from disjoint terms. In (J25), the remaining part of the same level is exactly

$$D_Y^2 \mathcal{H}_{N,1} + \nu_Y [\mathcal{H}_{N,1} - h(N/t^2)],$$

integrated against dt/t . In (J23), the remaining part is the full second level outside the displayed terminal interval. Both descriptions preserve the entire signed expression.

73. Checks, the surviving signed comparison, and updated status

The derivative responses $\mathcal{H}_{N,r}$ retain signs. Even-coefficient positivity therefore does not make every level positive. This can be checked analytically: at $N = 64$, for

$$8/\sqrt{3} < t < 4\sqrt{2},$$

the argument $64/t^2$ lies between two and three, only $d = 1$ occurs, and its continuous child vanishes. Hence $\mathcal{H}_{64,1}(t) = -1$ on this whole interval. The matching positive coefficient must remain coupled to the other intervals and levels, exactly as in (J19).

The new diagnostic script evaluates the derivative integrals and the original odd Möbius tent sum separately. It includes square boundaries $N = 9, 25, 49, 81$, a cube boundary $N = 27$, and ordinary dyadic cutoffs. These computations check conventions and identities; the proofs above are analytical.

N	Direct residual	First level	Second level	All higher levels combined
8	-0.198395	-0.305781	0.107386	0.000000
16	0.702913	0.390018	0.320717	-0.007822
32	1.207232	1.062697	0.159632	-0.015096
64	1.312628	1.916836	-0.658812	0.054604
128	1.486366	2.336627	-0.549572	-0.300690

In this run the full unanchored reconstruction discrepancy was below $1.3 \cdot 10^{-8}$. The trace identity at $\sqrt{N}$ and prime-jump identity agreed to floating-point precision. Additional tests reset the baseline at $Y = 3$, $Y = 5$, $Y = \frac{1}{2} \log 128$, and the sharper moving cutoff at $N = 512$, including baselines that are themselves primes. The largest discrepancy for the full reset formula, with the small interval added back, was below $7 \cdot 10^{-8}$. These are non-certified floating-point diagnostics and imply no estimate at untested scales.

The source package's diagnostics/ folder contains:

- `joint_transfer_diagnostics.py`, the editable calculation;
- `joint_transfer_diagnostics.json`, the complete recorded results.

The direct attempt has established:

Result	Location	Status
Regularity and matching square-root traces	(J2)-(J6)	Proved
Prime jumps equal inherited smaller responses	(J7)-(J8)	Proved
Shared boundary cancellation with post-jump correction	(J9)-(J10)	Proved
Complete finite expansion at any baseline	(J11)-(J15)	Proved
Full small-insertion interval fits the permitted rate	(J16)-(J19)	Proved
All even coefficient polynomials are nonnegative	(J20)-(J22)	Proved
Terminal square credit and whole unit-tent prime-square credit	(J23)-(J26)	Proved, with distinct grouping rules
Original all-scale lower bound for the complete residual	(Z14), equivalently the signed integral below	Remains unproved here

After the paid interval, the exact sufficient comparison is

$$\int_{Y_N}^N \left[\sum_{k=1}^{q_{Y_N}-1} k \mathcal{B}_{k,Y_N}(t) \mathcal{H}_{N,k-1}(t) \right] \frac{dt}{t} \geq -C_\varepsilon N^{1/2+\varepsilon}. \quad (\text{J27})$$

This is the same original one-sided bound with an explicitly bounded part removed. No square-budget law, separate energy condition, or additional analytic closing premise has been introduced. Equation (J19) returns it directly to $\mathcal{Z}_N$, and section 62 supplies the conditional zero-exclusion argument.

The next signed comparison must retain the discrepancy powers, prime counts, and inherited derivative responses together. The formulas above identify actual positive credits and pay a complete family, but they do not establish that these credits dominate all negative contributions in (J27).